A fixed Diophantine representation of Con(ZFC)

Prepared for Vladimir Reshetnikov's ProveIt repository
Mathematical construction and reproducible arithmetic artifacts

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Abstract

We give an explicit system of seven ordinary polynomial equations in sixteen strictly positive integer unknowns whose lack of solutions is equivalent, provably in ZFC, to Con(ZFC). Its largest equation degree is 28; a shared arithmetic circuit uses 63 binary operations. The same construction gives a single polynomial of degree 101 in 74 operations, or a single polynomial of degree 157 in 72 operations. A separate explicit lift gives 70 quadratic equations in 79 nonnegative unknowns and a single quartic with the same solvability predicate.

Every coefficient is fixed by a terminating finite recipe applied to the literal K215 transition table. No unspecified universal-machine index is left over. The coefficients are enormous and are specified exactly in compressed form, not expanded into decimal numerals. The new contribution is a mathematical proof and executable source audit, using the repository's existing K215 calibration and Diophantine compiler theorems. It is not a new end-to-end Lean or Rocq formalization. No global minimality is claimed.

Contents

1 The recommended system 2

2 Exact fixed coefficients 4

3 Proof of the arithmetic equivalence 7

4 From K215 to consistency in ZFC 11

5 Exact degree and source counts 12

6 Domains and the explicit quartic alternative 12

7 Reproducibility, evidence and limitations 13

A Source interface and reproducible commands 14

B Pinned mathematical dependencies 15

1 The recommended system

All five coefficient ports below denote fixed integers:

$$B-1, \quad K, \quad M_C, \quad \overline{M}_F, \quad W_0.$$

Their complete specification is in Section 2; in particular,

$$b = 5^{43}, \qquad d = 5^{87}, \qquad B = 2^d, \qquad W_0 = 2^{2d+b}. \tag{1}$$

The other three coefficient ports encode a single explicitly specified finite local rule set obtained from K215. They are not unknowns.

Take precisely sixteen strictly positive integer unknowns, in this order:

$$J, F, \alpha, t, f, h, i, T, s, w, \tau, \eta, \zeta, y, Z, \gamma. \tag{2}$$

Define the following polynomial abbreviations. They add no existential variables and use no variable exponentiation:

$$\begin{aligned}
q &= (B-1)J+1, & X &= wq, & Y &= sq^3,\\
E &= XY, & k &= \eta+\zeta, & c &= kY+\eta,\\
a &= E+Y, & H &= 4a+3, & \Delta &= a^2+H,\\
D &= X+ac+\gamma H, & C &= Z+W_0, & G &= q^2-qF-Z,\\
R &= G(q^2-1)+(M_C+q\overline{M}_F)J, & & & &\\
V &= c(Tf-1)-Rf^2, & L &= E(kY), & &\\
I &= ic^2, & K_a &= \Delta(f^2-1). & &
\end{aligned} \tag{3}$$

The recommended system $\mathcal{S}$ is

$$\boxed{\begin{aligned}
\tau^2 - L(L+k) &= 1,\\
D^2 - \Delta c^2 &= 1,\\
K_a(V^2-y^2)+y^2 &= 1,\\
k - hE - R &= 1,\\
(K+w)C + (q-F) - t(q-1) &= 1,\\
I^2 &= K_a,\\
q - F - Z - \alpha &= C.
\end{aligned}} \tag{4}$$

For example, the sixth equation is exactly $(ic^2)^2 = \Delta(f^2-1)$. It does not impose $\Delta \mid i$. The source file `fixed_input63_ordinary_system.json` gives every arithmetic instruction and the seven comparisons. Its witness domain is part of the specification: allowing arbitrary signed values in (2) without a domain conversion is not the same problem.

Theorem 1.1 (Fixed ZFC instance)**.** *For the coefficients of Section 2, ZFC proves*

$$(\exists \boldsymbol{x} \in \mathbb{N}_{>0}^{16}\ \mathcal{S}(\boldsymbol{x})) \quad \longleftrightarrow \quad \neg\,\mathrm{Con}(\mathsf{ZFC}).$$

Consequently ZFC proves

$$\boxed{(\nexists \boldsymbol{x} \in \mathbb{N}_{>0}^{16}\ \mathcal{S}(\boldsymbol{x})) \quad \longleftrightarrow \quad \mathrm{Con}(\mathsf{ZFC}).}$$

Here $\mathrm{Con}(\mathsf{ZFC})$ *is the usual arithmetical assertion that no finite ZFC derivation proves a contradiction, with the proof calculus calibrated by the pinned K215 endpoint.*

1.1 Single-polynomial versions

First define two parent certificates whose unit-sign proofs are especially transparent. One final simplification below removes a gate from each. Write

$$\begin{aligned} N_0 &= \tau^2 - L(L+k), & N_1 &= D^2 - \Delta c^2, \\ N_a &= K_a(V^2 - y^2) + y^2, & N_k &= k - hE - R, \\ N_t &= (K+w)C + (q-F) - t(q-1), & N_s &= 1 + I^2 - K_a, \\ M &= 1 + (q - F - Z - \alpha - C)^2. \end{aligned}$$

Then a complete single equation is

$$\boxed{P_{75} := N_0 N_1 N_a N_k N_t N_s M - 1 = 0.} \tag{5}$$

It has exactly the same positive solution set as (4). The reverse implication is not a generic consequence of taking products: Section 3 proves that all possibly negative unit factors have the required positive signs.

There is a slightly shorter circuit if higher degree is acceptable. Keep the common abbreviations, but set

$$\widetilde{K}_a = (i\Delta c^2)^2, \quad \widetilde{N}_a = \widetilde{K}_a(V^2 - y^2) + y^2, \quad \widetilde{N}_s = \Delta f^2 - \widetilde{K}_a.$$

The alternative equation is

$$\boxed{P_{73} := N_0 N_1 \widetilde{N}_a N_k N_t \widetilde{N}_s M - \Delta = 0.} \tag{6}$$

It has a positive solution if and only if (4) has one. The solution sets are not asserted to coincide with the same i: a solution of (6) gives an ordinary-strong solution by $i_{\text{ordinary}} = \Delta i_{\text{normalized}}$. Soundness in the other direction is proved directly, not by dividing an arbitrary i by Δ.

For the final shortest circuits, put

$$r_0 = q - F - Z - \alpha - C$$

and remove the addition of 1 from the marker factor. The resulting single equations are

$$\boxed{P_{74} := N_0 N_1 N_a N_k N_t N_s \, r_0^2 - 1 = 0,} \tag{7}$$

$$\boxed{P_{72} := N_0 N_1 \widetilde{N}_a N_k N_t \widetilde{N}_s \, r_0^2 - \Delta = 0.} \tag{8}$$

Their positive solvability is equivalent to that of (4). Their individual positive zero tuples need not satisfy the old budget equality: now $r_0 = \pm 1$. The precise soundness and completeness argument is in Section 3.6. The seven-equation source remains unchanged.

A direct sum-of-squares polynomial, often easiest to trust mechanically, is

$$P_{\text{SOS}} = \sum_{j=1}^{7} r_j^2, \tag{9}$$

where r_j is the left side minus the right side of the jth equation in (4). Integer solvability of $P_{\text{SOS}} = 0$ with the specified positive primary domain is exactly system solvability.

1.2 What "simplest" means here

There is no single agreed ordering of variable count, degree, coefficient size, expanded term count, formula length and shared arithmetic cost. The seven-equation system is our recommendation because it is short to read, has sixteen unknowns and exposes the proof obligations directly. The alternatives make the tradeoffs explicit.

Presentation	Unknowns	Equations	Degree	Operations
Recommended system $\mathcal{S}$	16 positive	7	28 max.	63
P_{74}	16 positive	1	101	74
P_{72}	16 positive	1	157	72
P_{SOS}	16 positive	1	56	83
Integer SOS	64 integers	1	112	211
Quadratic lift	79 nonnegative	70	2 max.	—
Quartic lift	79 nonnegative	1	4	—

Table 1: Exact counts for the supplied presentations. The circuit count charges each binary addition, subtraction and multiplication, including one multiplication per square and arithmetic involving fixed numerals. Reuse and comparisons are free. Coefficient bit lengths are not charged.

The pinned repository also proves the existence of a nine-unknown representation for every recursively enumerable set, through `Jones1982.nine_unknowns` in `NineMain.lean` [10]. Specializing that theorem to the inconsistency predicate gives a nine-unknown existence result. It does not make the displayed sixteen variables minimal, nor does it supply a smaller literal coefficient packet in this contribution. Our target is an inspectable concrete equation tied to the literal K215 machine, with short source and checked arithmetic transformations.

2 Exact fixed coefficients

2.1 The finite machine underlying every coefficient

The source is the literal binary table `Computability/BusyBeaver/Research/ZFC/candidate-k215/k215.tm` at the pinned snapshot [1]. The supplied copy has SHA-256

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090.`

It has 430 transition rows, initial state 0, working states 0 through 214, halt state 215, and blank symbol 0 on a two-way tape. Its only transition into the halt state is `68 1 1 R 215`.

The explicit program `normalized_k215.py` turns this into a 436-state, eight-symbol, one-sided machine $\mathcal{M}$ with a stationary first step and absorbing acceptance. Physical cell $-j$ stores the original cells j and $-j-1$ as two bits. An origin flag marks cell 0. The eight tape symbols encode

$$(0,0,0),(1,0,0),(0,0,1),(0,1,0),(1,1,0),(0,1,1),(1,0,1),(1,1,1)$$

in that order. States 0 through 5 are START, AFTER, ERASE, RETURN, SINK and HALT. State $6+2q+e$, for $0 \le q < 215$ and $e \in \{0,1\}$, simulates original state q on the nonnegative or negative half of the original tape. All 3,488 transitions are explicitly defined.

An input of $n \ge 3$ ones is erased, the head returns to the marked origin, and blank K215 simulation begins after $2n+2$ initialization steps. A virtual move across $0/-1$ switches the selected track while staying at the physical origin. Other moves follow the folded tape. The tape and head correspondence is an inductive invariant, proving for every such input that $\mathcal{M}$ accepts if and only if K215 halts blank. This conclusion is an all-time simulation proof, not an inference from the finite execution tests.

The finite window compiler uses raw input 2 when its arithmetic input is fixed to $x = 1$. The normalized machine therefore starts with four ones. It deliberately erases its input, so this interface convention does not affect the desired halting predicate. It would be incorrect to claim that this particular machine recognizes only the even inputs.

2.2 The ordered window list

The tile alphabet has size

$$a_{\text{tile}} = 2 + 8(436 + 1) + 4 = 3502.$$

IDs 0 and 1 are the vertical and horizontal boundary tiles. They are followed by the payload tiles (symbol, head), in lexicographic order with head values None, $0, \ldots, 435$, followed by the four initial phases L, I, Q, R. The exact local validity test is the total finite predicate `local_valid` in the supplied source. It enforces boundary propagation, the phase grammar LL, LI, II, IQ, QR, RR, one-step tape updates, no escaping head, and acceptance on a terminal row.

Place the following two row-major 3×3 windows first:

$$w_0 = (1, 1, 1, 3499, 3499, 3500, 439, 439, 878),$$
$$w_1 = (1, 1, 1, 3498, 3499, 3499, 2, 439, 439).$$

List all other locally valid nine-tuples in lexicographic order. This defines every w_j without reference to a run length or a halting oracle. The four corner coordinates do not affect the center predicate, but are retained in each window and in the overlap constraints.

The exact five-position cross count is 136449173881235, so the number of windows is

$$k_{\text{win}} = 136449173881235 \cdot 3502^4 = 205227463535357025370704197 60. \tag{10}$$

The formula, its derivation from transition counts, and independent small exhaustive counting controls are in `normalization.md` and `window_count_checks.json`. The actual enormous ordered list has not been expanded. Its direct iterator examines at most 3502^9 candidates, a fixed finite bound.

2.3 Clause layout

For the rest of this section put $a_t = 3502$ and $k_t = k_{\text{win}}$. Use distinct symbols for layout constants to avoid confusion with the polynomial abbreviations of Section 1:

$$A_t = 2^{95}, \qquad p_t = (k_t - 73736)/2,$$
$$j_\mu = 31522 + p_t = 10261373176767851268535204534,$$
$$\mu_t = A_t - 2 + \sum_{j=1}^{j_\mu} A_t^j,$$
$$m_t = k_t + 15a_t + 1 = 205227463535357025370704722 91.$$

For $r, c \in \{0, 1, 2\}$ and $0 \le s < a_t$ define

$$e(r, c, s) = k_t + 3a_t + (2 - r)3a_t + (2 - c)a_t + s, \qquad e_* = k_t + 12a_t.$$

The native position set is

$$\mathcal{E} = \{0, \ldots, k_t - 1\} \cup \{k_t + 3a_t, \ldots, k_t + 12a_t\} \cup \{m_t, 3m_t, 9m_t, 27m_t\}.$$

The final member of the middle interval is one ignored dummy position. Further define

$$\begin{aligned} h_t &= 51m_t + 3a_t + 1, & t_1 &= 105m_t + 4a_t + 2, \\ t_2 &= 159m_t + 4a_t + 3, & g_t &= 186m_t + 4a_t + 4, \\ \mathcal{L} &= 5^{44}, & b &= 5^{43}, \quad d = b\mathcal{L} = 5^{87}, \\ \mathcal{R} &= 2^b, & B &= 2^d. \end{aligned}$$

These are the compiler's literal least-power-of-five choices, not arbitrary upper bounds. In particular,

$$5^{43} < 213m_t + 4a_t + 5 \le 5^{44}, \qquad 5^{42} < 95j_\mu + 2 \le 5^{43}.$$

The detailed bit-length bound in `constants.md` proves that $95j_\mu + 2$ is the required radix bit length before taking the next power of five.

Each selector window contributes the fixed clause coefficient

$$c_j = 1 + \sum_{r,c=0}^{2} A_t^{1+(3r+c)a_t+(w_j)_{3r+c}} + \sum_{v=0}^{3} A_t^{1+9a_t+v} \quad (0 \le j < k_t).$$

For the copy positions and anchors use

$$c_{e(r,c,s)} = A_t^{1+(3r+c)a_t+s}, \qquad c_{\lambda_v m_t} = A_t^{1+9a_t+v}, \quad (\lambda_0, \lambda_1, \lambda_2, \lambda_3) = (1, 3, 9, 27).$$

Set $c_{e_*} = 0$. This defines the complete finite coefficient map.

2.4 The five actual coefficient ports

Define

$$\begin{aligned} D_C^0 &= \mathcal{R}^{3a_t} + \mathcal{R}^{h_t+a_t} + \mathcal{R}^{8m_t} + \mathcal{R}^{24m_t} \\ &\quad + \sum_{e \in \mathcal{E}\setminus\{e_*\}} c_e(\mathcal{R}^{t_1-e} + \mathcal{R}^{t_2-e}), \\ D_R &= \mathcal{R}^{h_t}, \\ \epsilon_t &= \begin{cases} 1, & 5 \mid 2D_C^0 - D_R, \\ 0, & 5 \nmid 2D_C^0 - D_R, \end{cases} \\ D_C &= D_C^0 + \epsilon_t \mathcal{R}^{g_t}, \qquad K = D_C + BD_R. \end{aligned}$$

The Boolean ϵ_t is a fixed integer determined by a finite expression, not an extra Diophantine variable. Its actual K215 value has not been evaluated; the formula fixes it uniquely. The correction ensures $2D_C - D_R \not\equiv 0 \pmod 5$.

The masks are

$$\begin{aligned} M_C &= B - 1 - \sum_{e \in \mathcal{E}\setminus\{1\}} \mathcal{R}^e - 2\mathcal{R}^{e_*}, \\ M_F^{\mathrm{nat}} &= 4 + \mu_t(\mathcal{R}^{t_1} + \mathcal{R}^{t_2}) \\ &\quad + \sum_{\substack{r=0,1;\ c=0,1,2 \\ 0 \le s < a_t}} \mathcal{R}^{e(r,c,s)} + \sum_{\substack{r=0,1,2;\ c=0,1 \\ 0 \le s < a_t}} \mathcal{R}^{h_t+e(r,c,s)} \\ &\quad + \mathcal{R}^{9m_t} + \mathcal{R}^{27m_t}, \\ \overline{M}_F &= M_F^{\mathrm{nat}} + B - 1, \qquad W_0 = \mathcal{R}B^2 = 2^{2d+b}. \end{aligned}$$

The native masks satisfy

$$0 < M_C, M_F^{\mathrm{nat}} < B - 1, \qquad M_C \equiv 2 \pmod 4, \quad M_F^{\mathrm{nat}} \equiv 4 \pmod 8,$$
$$\mathrm{popcount}(M_C) + \mathrm{popcount}(M_F^{\mathrm{nat}}) = d.$$

The polynomial source name `MF` denotes $\overline{M}_F$, the *shifted* coefficient. Applying the native range bound to this shifted value would be an error. The source aliases are summarized in Table 2.

Source port	Exact mathematical value
`Bm1`	$B - 1$
`Kconstant`	$D_C + BD_R$
`MC`	M_C
`MF`	$M_F^{\mathrm{nat}} + B - 1$
`WO`	2^{2d+b}

Table 2: The five coefficient ports are fixed numerals, not parameters over which the theorem quantifies.

This is an ordinary polynomial specification: every exponent in this section is fixed independently of the sixteen unknowns. All coefficient computations have fixed finite bounds. No predicate such as "is a proof" or "halts" is hidden in a coefficient definition.

The coefficient representation is compact, but the integers are not small. B alone has $5^{87} + 1$ binary digits, approximately 6.46×10^{60} bits. We do not claim to have materialized these integers. `fixed_compiler.py`, `fixed_constants.json` and `constants.md` provide the exact algorithm and symbolic values.

3 Proof of the arithmetic equivalence

We prove that the fixed system and both compact single polynomials have a positive solution exactly when $\mathcal{M}$ accepts raw input 2. The proof uses the pinned fixed-program compiler theorem [3], its canonical completeness construction [4], and the specific Pell rank and step-down lemmas [5, 6, 7]. These are ordinary mathematical dependencies; the accompanying finite tests do not replace them.

For $A \geq 2$, define the Pell sequences by

$$\chi_A(n) + \psi_A(n)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^n \quad (n \geq 0).$$

Both sequences have integer values and satisfy their usual second-order recurrences. Every positive solution to $d^2 - (A^2 - 1)c^2 = 1$ is $(d, c) = (\chi_A(p), \psi_A(p))$ for some $p \geq 1$.

3.1 Recovering the unit signs

At a positive zero of P_{75}, its seven integer factors multiply to 1. Each is therefore 1 or -1. Since $M \geq 1$, it is 1, giving

$$q = F + 2Z + \alpha + W_0, \quad C = Z + W_0, \quad 0 < C < q, \quad F + Z < q, \quad W_0 < q. \tag{11}$$

Also $\Delta = (a + 2)^2 - 1$ is 0 or 3 modulo 4. The main norm N_1 cannot be -1 modulo 4. The ordinary strong factor is $N_s = 1 + I^2 - \Delta(f^2 - 1)$. If $\Delta \equiv 0$ it is $1 + I^2$ modulo 4; if $\Delta \equiv 3$ it is $I^2 + f^2$ modulo 4. Neither case can give 3. Hence $N_s = 1$ and $K_a = I^2 > 0$. Now N_a is congruent to y^2 or V^2 modulo 4, so $N_a = 1$.

For the first factor put $V_0 = XY^2 > 1$. Since $L = V_0 k$,

$$N_0 = \tau^2 - V_0(V_0+1)k^2.$$

The negative Pell equation for $V_0(V_0+1)$ has no integer solution. Indeed, suppose $x^2 - V_0(V_0+1)z^2 = -1$ with the smallest positive z. Then $V_0 z < x < (V_0 + \frac{1}{2})z$. Multiplication by the inverse unit $(2V_0+1) - 2\sqrt{V_0(V_0+1)}$ produces another solution with

$$0 < z' = (2V_0+1)z - 2x < z,$$

a contradiction. Thus $N_0 = 1$.

Write the two remaining signs as

$$N_k = \varepsilon, \quad N_t = \nu, \qquad \varepsilon, \nu \in \{-1, 1\}, \quad \varepsilon\nu = 1.$$

Their positivity will be proved, rather than assumed from the product.

3.2 Preliminary bounds and Pell indices

The actual native mask bounds, the shifted-mask identity and (11) give the pretyping estimates

$$(2q-1)(q^2-1) < R < q^4 - q^3, \quad E = wsq^4 \geq q^4 > R+2, \quad a > R+2, \tag{12}$$

with $q \geq B \geq 16$. They do not assume that q is dyadic or that a valid computation has already been decoded.

Put $A = a+2$ and $P_0 = 2XY^2 + 1$. The fundamental positive solution of the first Pell equation has $k = 2$, so

$$\tau = \chi_{P_0}(n), \qquad k = 2\psi_{P_0}(n), \qquad n \geq 1.$$

Since $P_0 \equiv 1 \pmod{E}$ and $k = R + \varepsilon + hE$,

$$2n \equiv R + \varepsilon \pmod{E}, \qquad n \geq (R-1)/2 \geq 24.$$

The main norm gives $D = \chi_A(p)$ and $c = \psi_A(p)$ for $p \geq 1$. As $P_0 > A$ and $c > k$, Pell monotonicity implies $p > n$. Elementary growth bounds now give

$$c > A\Delta^2, \qquad c > 2p, \qquad c > 2R. \tag{13}$$

These bounds precede every use of auxiliary positivity or strong rank.

The ordinary strong equality gives

$$f^2 = 1 + i^2 c^4/\Delta, \qquad f > c^2/\sqrt{\Delta} > c^2/A > 2c.$$

Thus $c > \Delta + 1$, $f^2 > \Delta + c$, and $R < a < \Delta$ imply

$$K_a - (Rf^2 + c) = (\Delta - R)f^2 - \Delta - c > 0. \tag{14}$$

Using the stronger normalized-source bound $f > c^2$ here would be unjustified; (13) is what makes the ordinary version work.

3.3 Restoring positive auxiliary witnesses

The auxiliary norm is

$$K_a V^2 - (K_a - 1)y^2 = 1.$$

If $v = |V| > 1$, then $v^2 - 1 = (K_a - 1)(y^2 - v^2)$, whence $y \geq v + 1$ and $v \geq 2K_a - 1$. But $T > 0$ and (14) give

$$V = c(Tf - 1) - Rf^2 > -Rf^2 - c > -K_a.$$

This excludes negative V with $|V| > 1$. The cases $V = 0, 1, -1$ are excluded by the norm or by $V \equiv -c \pmod{f}$ and $2 < c < f/2$. Therefore $V > 1$ and $y > V$.

The generic strong-rank lemma [5] applies to $c = \psi_A(p) > A\Delta^2$ and $(ic^2)^2 = \Delta(f^2 - 1)$. It supplies an index m with

$$f = \chi_A(m), \quad p \mid m, \quad c \mid m, \quad ic^2 = \Delta\psi_A(m), \quad m \geq c > 2p. \tag{15}$$

In particular,

$$f^2 - 1 = \Delta\psi_A(m)^2 = ic^2\psi_A(m),$$

so $c^2 \mid f^2 - 1$. This integrality fact is essential. Define

$$o = cT - Rf, \qquad j = Tf - 1 - \frac{R(f^2 - 1)}{c}. \tag{16}$$

They are integers and satisfy $of - c = V$ and $jc = V + R$. Since the right sides are positive, $o, j > 0$.

The signed auxiliary step-down lemma [6] now has precisely its required hypotheses: (15), $V = of - c = jc - R > 0$, the auxiliary norm, and $0 < p, R < c/2$. It yields $p = R$. Its proof classifies the auxiliary Pell solution at an odd index, uses its congruences modulo f and c, then excludes the negative and wrapped target alternatives using these strict bounds.

Write $2n = R + \varepsilon + vE$. Because $0 < R + \varepsilon < E$, $v \geq 0$. If $v \geq 1$, then $2n \geq R - 1 + E > 2R + 1$, contradicting $n < p = R$. Therefore $2n = R + \varepsilon$. If $\varepsilon = -1$ then $p = 2n + 1$. Pell duplication and monotonicity give

$$c = \psi_A(2n + 1) > \psi_A(2n) \geq Ak > k(Y + 1),$$

contrary to $kY < c < k(Y + 1)$, which follows from $\eta, \zeta > 0$. Hence $\varepsilon = \nu = 1$. Every equation of (4) has now been recovered from $P_{75} = 0$. The converse follows by direct multiplication.

3.4 Recovering the power and the omitted fixed-input bridge

Now $R = 2n - 1$ is odd. Write $R = 2r + 1$. The input-independent asymmetric Pell-ratio argument [7] gives

$$Y > \frac{(X + 1)^{2r}}{2X^r} - 1 > \frac{X^r}{3}, \qquad a > \frac{X^{r+1}}{3} > 2^R.$$

For $E_A(v) = \chi_A(v) - a\psi_A(v)$, the initial values are 1 and 2, and its recurrence gives $E_A(v) \equiv 2^v \pmod{H}$. The main root says $E_A(R) = X + \gamma H$. Since both X and 2^R are strictly between 0 and $a < H$, their congruence forces

$$X = 2^R. \tag{17}$$

Because $X = wq$, q is a power of two, say $q = 2^\ell$. The lower bound in (12) gives $R > 3q > 3\ell$, so $q^3 \mid X$ and $w_{\text{raw}} = X/q^3 > 0$. Also $(B - 1) \mid (q - 1)$ implies $q = B^N$ for some $N \geq 1$.

Set $u = 2d + b$, so $W_0 = 2^u$. The budget yields $u < \ell < R$. The fixed-input replacement omitted the old subtraction $2d$ from the budget; its positive restoration must be justified. Since q and W_0 are distinct powers of two, $q \geq 2W_0$, and

$$q - C = \frac{q - W_0 + F + \alpha}{2} > \frac{W_0}{2} \geq 2d. \tag{18}$$

Consequently $\alpha_{\mathrm{raw}} = q - C - 2d > 0$ is a valid raw slack. The tempting formula $\alpha - 2d$ need not be positive and is not used.

Restore the deleted input variables by

$$\begin{aligned} \kappa &= \psi_A(u), & \mu &= \chi_A(u), \\ \delta &= (\kappa - u)/\Delta, & \phi &= c - \kappa, \\ \rho &= (E_A(u) - W_0)/H. \end{aligned}$$

All five are positive integers. Since u is odd, $\psi_A(u) \equiv u \pmod{\Delta}$, and growth gives $\psi_A(u) > u$. As $u < R$, $\phi > 0$. The projected recurrence proves ρ integral, and $E_A(u) > 2^u$ proves $\rho > 0$. They satisfy exactly

$$\begin{aligned} \kappa &= u + \delta\Delta, & c &= \kappa + \phi, \\ \mu^2 &= 1 + \Delta\kappa^2, & \mu &= W_0 + a\kappa + \rho H. \end{aligned}$$

Finally let $z_{\mathrm{raw}} = t - 1 + wC > 0$. The transport equation gives

$$(K + X)C - F = (t - 1 + wC)(q - 1).$$

Together with (16), $i_{\mathrm{raw}} = i$, w_{raw} and α_{raw}, these are all nineteen equations of the original raw75 certificate at arithmetic input 1. Every restored coordinate is positive. Its fixed-program compiler theorem therefore proves acceptance of raw input 2.

3.5 Completeness and the 73-operation alternative

If $\mathcal{M}$ accepts raw input 2, the pinned canonical F84 construction [4] provides its eighteen positive coordinates at arithmetic input 1. Its input bridge has $W = W_0$. Discard δ, ρ, σ, set

$$\gamma = \rho + \sigma, \qquad \alpha_{\mathrm{new}} = \alpha_{\mathrm{F84}} + 2d,$$

and keep the other coordinates. The resulting sixteen-tuple satisfies the normalized equations and $P_{73} = 0$. In that polynomial,

$$\widetilde{N}_s = \Delta(f^2 - \Delta i^2 c^4), \qquad \Delta > 0.$$

Canceling Δ turns its product equation into a product of integer units. The same sign and index argument works, with the simpler strong bound $f^2 = 1 + \Delta i^2 c^4$. Its full proof is supplied in `FIXED_INPUT73_PROOF.md`.

Map this normalized solution to the ordinary one by $i_{\mathrm{ordinary}} = \Delta i_{\mathrm{normalized}}$. Then

$$\Delta(f^2 - 1) = (i_{\mathrm{ordinary}} c^2)^2 = (i_{\mathrm{normalized}} \Delta c^2)^2,$$

so the auxiliary coefficient and all other required equations agree. This proves completeness of (4) and P_{75}. Both compact polynomials therefore represent the same fixed acceptance predicate.

3.6 The final squared-marker saving

At a zero of P_{74}, or of P_{72} after canceling $\Delta > 0$, the square factor r_0^2 is an integer unit. Hence $r_0 = \epsilon$ with $\epsilon \in \{-1, 1\}$. The new budget is

$$q = W_0 + F + 2Z + \alpha + \epsilon.$$

Since $\alpha + \epsilon \geq 0$, it still implies $C < q$, $F + Z < q$ and $W_0 < q$. These are all the budget consequences used before dyadic power recovery. The norm signs, first and main Pell indices, auxiliary positivity, rank, signed target recovery and no-wrap argument therefore remain valid.

Once q is known to be a power of two, $q \geq 2W_0$ and

$$q - C = \frac{q - W_0 + F + \alpha + \epsilon}{2} > \frac{W_0}{2} \geq 2d,$$

because $F + \alpha + \epsilon \geq 1$. Thus the original strictly positive slack $\alpha_{\text{raw}} = q - C - 2d$ is still restored. All other restorations are unchanged. This proves soundness directly through the full raw75 compiler, including the case $\alpha = 1$, $\epsilon = -1$.

For completeness, take a positive zero of P_{75} or P_{73} and replace α by $\alpha + 1$. Its old budget residual was zero; its new residual is -1, with square 1. No other factor uses α, so this gives a zero of P_{74} or P_{72}. The equivalence concerns existence of positive zeros. A pointwise inverse using $\alpha + \epsilon$ would incorrectly allow a zero coordinate in the edge case just mentioned.

4 From K215 to consistency in ZFC

The pinned Lean endpoint is

```
KEndpoint215.blank_halts_iff_zfc_refutation :
  BlankHalts machine215 <-> BProv ZFCax_s [] Form.fBot
```

The Rocq development gives the corresponding inconsistency endpoint. The upstream verification README distinguishes these external proof-assistant theorems from a proof encoded inside the object's own ZFC syntax [1, 2]. Its reported Lean axioms are `propext`, `Classical.choice` and `Quot.sound`; the Rocq endpoint reports are closed. These are upstream records, not fresh builds performed for this packet.

The ordinary mathematical calibration is formalizable in ZFC. Its numerical derivability predicate is inductively generated over the natural numbers; the set-theoretic calibration uses first-order completeness, set models, relative constructibility, reflection and quotient constructions. The simulation, finite-window compiler, Pell arguments and coefficient computations are ordinary finite combinatorics and arithmetic. No proper-class model or inaccessible universe is needed to transfer this chain to ZFC.

Combining the three equivalences gives

$$\exists \boldsymbol{x}\, \mathcal{S}(\boldsymbol{x}) \iff \mathcal{M} \text{ accepts raw input } 2 \iff K215 \text{ halts blank} \iff \neg\,\mathrm{Con}(\mathsf{ZFC}).$$

Each equivalence, including the finite definition of the coefficients, is provable in ZFC. Negating yields Theorem 1.1.

This is an equivalence theorem, not a proof of consistency. If ZFC is consistent, the constructed equations have no solutions in the standard integers, and ZFC cannot prove that lack of solutions by Gödel's second incompleteness theorem. Bare consistency alone should not be advertised as proving independence in both directions: excluding a ZFC proof of the false Σ_1 assertion of a solution requires an additional soundness assumption, such as Σ_1-soundness.

5 Exact degree and source counts

The seven-equation circuit has 34 multiplications and 29 additions or subtractions. All sixteen unknowns, all five coefficient ports and all 63 gates are live. The exact equation degrees, in the order of (4), are

$$22,\ 18,\ 28,\ 7,\ 2,\ 22,\ 1.$$

The apparent degree-22 leading terms in the main norm cancel; its true degree is 18. Counts based solely on syntactic degree propagation would therefore be inaccurate.

Forming the seven residuals, squaring them and adding them uses another $7+7+6 = 20$ operations, giving 83 for P_{SOS}. Its exact degree is 56 because the auxiliary residual is the unique one of degree 28.

For P_{75} the factor degrees are $22, 18, 28, 7, 2, 22, 2$, summing to 101. Put $Q = (B-1)J$ and $k_0 = \eta + \zeta$. Its top homogeneous form is

$$8Q^{55}h\gamma i^2k_0^9w^7s^{16}T^2f^4(wZ - tQ)(Q - F - 2Z - \alpha)^2.$$

The monomial $J^{58}h\gamma i^2\eta^9w^7s^{16}tT^2f^4$ has coefficient $-8(B-1)^{58} \neq 0$. Thus degree 101 is exact on every admissible coefficient slice. The circuit ledger is $41M + 34A = 75$.

For P_{73} the factor degrees are $22, 18, 60, 7, 2, 46, 2$, summing to 157. Its top homogeneous form is

$$-8Q^{91}h\gamma i^4k_0^{13}w^{13}s^{26}T^2f^2(wZ - tQ)(Q - F - 2Z - \alpha)^2.$$

The monomial $J^{94}h\gamma i^4\eta^{13}w^{13}s^{26}tT^2f^2$ has coefficient $8(B-1)^{94} \neq 0$. Subtracting Δ, of degree 12, cannot affect it. The ledger is $41M + 32A = 73$.

The final squared-marker versions delete exactly one addition each. Their ledgers are therefore $41M + 33A = 74$ and $41M + 31A = 72$. Both marker factors have the same leading homogeneous form r_0^2, so the exact degrees remain 101 and 157. The old 75- and 73-operation sources are retained as proof parents and arithmetic controls.

The reduction from the universal F84 source removes a variable-input Pell bridge that is unnecessary once arithmetic input 1 is fixed. The fixed marker W_0 replaces its role, and the positive main coordinate $\gamma = \rho + \sigma$ replaces two coordinates. The raw slack restoration in (18) permits a further subtraction to disappear. This is why the result is specifically a fixed predicate, rather than a new ordinary-input universal circuit with the same counts.

6 Domains and the explicit quartic alternative

6.1 Nonnegative and unrestricted integer domains

For a standard nonnegative domain $\mathbb{N} = \{0, 1, 2, \ldots\}$, replace each positive primary variable x_j by $1 + v_j$, $v_j \in \mathbb{N}$. This preserves the variable count, polynomial degree and solvability. The compact operation counts in Table 1 refer to the original positive-input interfaces; shifts are not silently free additional circuit instructions.

For an equation over unrestricted integers, Lagrange's four-square theorem gives a direct explicit conversion. For each $j = 1, \ldots, 16$ put

$$x_j = 1 + u_{j1}^2 + u_{j2}^2 + u_{j3}^2 + u_{j4}^2, \qquad u_{jr} \in \mathbb{Z}.$$

Substitute these expressions into P_{SOS}. The resulting single ordinary polynomial in 64 unrestricted integer unknowns has degree 112 and has a zero if and only if $\mathcal{S}$ has a positive solution. Equivalently, substitute into the seven equations to get a system with maximum degree 56. The construction and degrees are explicit; no positivity predicate remains outside the polynomial. The complete 211-operation circuit is supplied in `integer64_sos.json`; `emit_sos_variants.py` regenerates it and the 83-operation positive-domain SOS source.

6.2 A smaller-degree lift over the nonnegative integers

The circuit of $\mathcal{S}$ has 63 gates, but a naive lift cannot automatically make all their values nonnegative: subtraction gates need not be positive on arbitrary assignments. Here a solution-specific audit succeeds after one exact sign transformation.

Replace the three auxiliary instructions

$$G_a = V^2 - y^2, \quad H_a = K_a G_a, \quad N_a = H_a + y^2$$

by

$$G_a = y^2 - V^2, \quad H_a = K_a G_a, \quad N_a = y^2 - H_a.$$

The final auxiliary polynomial is identically unchanged, while two internal registers change sign. Section 3 proves $y > V > 1$ at every positive solution. The complete audit in `QUADRATIC_LIFT.md` checks every subtraction gate; all 63 modified gate values are nonnegative at such a solution.

Introduce $v_1, \ldots, v_{16} \in \mathbb{N}$ with $x_j = 1 + v_j$, and one nonnegative unknown g_r for each gate. An instruction $r = u \circ v$ becomes

$$g_r - \widehat{u} \circ \widehat{v} = 0, \qquad \circ \in \{+, -, \cdot\},$$

where each operand is a fixed coefficient, an affine primary expression $1 + v_j$, or an earlier gate variable. The resulting 63 gate equations have degree at most two. Add the seven final comparisons. This gives 79 nonnegative unknowns and 70 quadratic equations.

The forward implication uses the proven nonnegativity of each gate at a solution. Conversely, topological induction on the gate equations recovers the evaluated circuit and then every equation of $\mathcal{S}$. Thus no extra or lost solutions are introduced.

If $E_1, \ldots, E_{70}$ are the explicitly emitted residuals, then

$$Q_{79} = \sum_{r=1}^{70} E_r^2 = 0 \tag{19}$$

is a single polynomial of degree four with the same solvability predicate over $\mathbb{N}^{79}$. The first gate equation is $g_{\text{tau_square}} - (1 + v_{\text{tau_root}})^2 = 0$. Its square supplies the coefficient $[v_{\text{tau_root}}^4]Q_{79} = 1$, proving exact degree four. `quadratic_lift.json` contains all 70 residuals and the entire expanded sparse quartic: 286 formal terms before the five fixed coefficient ports are specialized. The ports remain coefficients, not extra unknowns.

7 Reproducibility, evidence and limitations

The archive contains the complete LaTeX source and PDF, standalone Python programs using only the standard library, literal arithmetic circuits, an expanded quartic, exact coefficient definitions, the K215 table, proof notes, review notes and machine-readable receipts. The `README.md` gives the file map and commands.

The arithmetic validators authenticate the parent F84 JSON by its Git blob hash, reconstruct the child row changes, check closure and liveness, recount every operation, compare signed full-source identities, and verify exact degree by integer polynomial arithmetic. The ordinary variant also checks the strong-coordinate transformation. These checks use small diagnostic coefficient tuples only as algebraic controls; those tuples are not claimed to encode ZFC.

The normalized machine checks all 3,488 rows, 5,160 folded transition cases, four input erasures, and matched finite simulation prefixes. The exact window-count formula was checked against exhaustive five-position enumeration on two small machines. The finite compiler was compared

field by field with the two pinned reference compiler functions on five fixtures, covering both correction-bit branches. On the smallest fixture the complete fixed numerals were materialized, and native mask ranges, populations and modular evaluations checked.

The quartic generator proves the local sign identity by sparse polynomial arithmetic, emits all residuals, expands their squares, checks the exact degree and coefficient, and compares the result on 64 signed circuit assignments. Separate reviews rederived the main sign and index arguments, exact degree leaders and coefficient layout. These are reviews and tests produced within this work, not a claim of independent human peer review.

What has not been computed or formalized. The actual K215 window list, expanded giant coefficient tuple, correction bit and any accepting witness have not been materialized. No test decides K215 halting or ZFC consistency. The new end-to-end Diophantine theorem has not been compiled in Lean or Rocq, and the upstream proof assistants were not rerun here. The proof is an ordinary ZFC-formalizable mathematical proof with the explicit pinned dependencies listed below. This distinction is material when deciding where the packet belongs in the repository's research and verification directories.

A Source interface and reproducible commands

Mathematical variable	Literal source name
J	`Jrep`
F	`F`
α	`alpha`
t	`transport_quotient`
f	`f`
h	`h`
i	`i`
T	`auxiliary_quotient`
s	`s`
w	`w`
τ	`tau_root`
η	`eta`
ζ	`zeta`
y	`y_aux`
Z	`Z`
γ	`gamma`

From the extracted directory, run

```
python3 reproduce.py
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

The reproduction command performs the finite implementation and arithmetic checks. It never enumerates the actual huge window list or materializes the actual K215 coefficient tuple. The included JSON receipts record the scope of each check. The PDF build requires a normal TEX installation; the Python checks use no third-party packages.

B Pinned mathematical dependencies

Every ProveIt reference below is fixed to commit

`cdd582bec13f908df3711fa9e05d1ddac368a5db.`

The repository base is https://github.com/VladimirReshetnikov/ProveIt. The full paths in the bibliography can be appended to its `blob/<commit>/` route. This avoids an evolving `main` branch changing the meaning of the cited theorem. The supplied `PROVENANCE.json` records source locations and hashes; `ATTRIBUTION.md` explains copied notices and inherited terms.

References

[1] V. Reshetnikov and ProveIt contributors. *K215 verification status and literal candidate.* `Computability/BusyBeaver/Research/ZFC/k225-formal/verification/k215/README.md`; `Computability/BusyBeaver/Research/ZFC/candidate-k215/README.md`; `Computability/BusyBeaver/Research/ZFC/candidate-k215/k215.tm`.

[2] ProveIt contributors. *K215 consistency calibration endpoint.* `Computability/BusyBeaver/Lean/BusyBeaver/ZFC/KEndpoint215.lean`, with its imported calibration and machine proofs.

[3] ProveIt contributors. *A 75-operation fixed-index universal certificate.* `Computability/HilbertTenthProblem/Papers/1980/FIXED_RAW_UNIVERSAL_75_PROOF.md`. This is the complete raw arithmetic/compiler theorem used for soundness.

[4] ProveIt contributors. *Scaled strong output, complete F84 source.* `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete84_scaled_strong_output.md` and the adjacent `.json` source. The source JSON Git blob is `a8e6e50e96503e69221b8586e273208e5345eee4`.

[5] ProveIt contributors. *Relaxed auxiliary Pell proof.* `Computability/HilbertTenthProblem/Papers/1980/PELL_RELAXED_AUXILIARY_PROOF.md`. See also `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_coupled_index_linear88.md` for the generic strong-rank application before index recovery.

[6] ProveIt contributors. *Reversed auxiliary construction.* `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_reversed_auxiliary89.md`, especially the signed index step-down in Section 4.

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[9] ProveIt contributors. *Finite helical window compiler and modified masks.* `Computability/HilbertTenthProblem/Papers/verification/explore_fixed_raw_universal_76.py`; `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_half_binomial_compiler.py` and its adjacent explanatory `.md` file.

[10] ProveIt contributors. *Nine-unknown representation theorem.* `Computability/HilbertTenthProblem/Lean/Diophantine/Paper1982/NineMain.lean`, theorem `Jones1982.nine_unknowns`.

[11] J. P. Jones. Universal Diophantine equation. *Journal of Symbolic Logic* 47 (1982), 549–571. https://www.cambridge.org/core/journals/journal-of-symbolic-logic/article/abs/universal-diophantine-equation/113EBF3D928DFC6011C52BECAE81A087.

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